COMPREHENSIVE CYBERSECURITY AND ETHICAL HACKING PRACTICAL COURSE

DURATION: 9 Months (3 Phases)

Phase 1: Basic Cybersecurity Concepts (3 Months)

Month 1: Foundations of Cybersecurity, Linux, and Windows Basics

Week 1: Introduction to Cybersecurity

- Overview of cybersecurity principles: CIA triad, threats, and vulnerabilities.
- Cybersecurity domains: Network, Web, Application, Cloud, and IoT.
- Types of cyberattacks: Malware, phishing, DDoS, ransomware.
- Cybersecurity tools overview (Wireshark, Nmap, Metasploit).

Week 2: Linux Basics and Essentials

- Installing Linux (Ubuntu and Kali/Parrot) on VirtualBox/VMware.
- Basic Linux commands: Navigation, file management, and permissions.
- Linux file system hierarchy and shell basics.
- Practical: Navigate, create, modify, and secure files/directories in Linux.

Week 3: Windows OS Basics

- Understanding the Windows architecture.
- Overview of system environment variables and modifying PATH variables.
- Windows file systems (FAT32, NTFS) and folder structures.
- Practical: Setting up and customizing user profiles, modifying PATH variables.

Week 4: User Management on Linux and Windows

- Linux: Adding, modifying, and deleting users and groups.
- Windows: Local users, groups, and permissions (User Account Control).
- Configuring user rights assignments and enforcing security policies.
- Practical: Manage users and permissions on both Linux and Windows.

Month 2: Networking and System Security Basics

Week 1: Networking Fundamentals

- OSI and TCP/IP models.
- IP addressing, subnetting, and NAT.
- Tools: ping, traceroute, netstat, and Wireshark basics.
- Practical: Analyze packets using Wireshark and simulate basic networks with VirtualBox.

Week 2: Web and Internet Architecture

- DNS, HTTP/HTTPS, web servers (Apache, NGINX, IIS).
- How websites work: DNS resolution and HTTP requests.
- Hands-on: Setting up a simple web server on Linux and Windows.

Week 3: Windows System Administration Essentials

- Task Manager and Resource Monitor.
- Managing services and startup programs using MSConfig.
- Windows registry basics and security settings.
- Practical: Analyze and secure system processes, services, and registry settings.

Week 4: Cryptography Basics

- Introduction to encryption, hashing, and digital certificates.
- Symmetric vs. asymmetric encryption.
- Practical: Encrypting files with GPG and managing SSL/TLS certificates.

Month 3: Governance, Security Tools, and Scripting Basics

Week 1: Governance, Risk, and Compliance (GRC)

- Overview of GRC and its importance in cybersecurity.
- Introduction to frameworks: NIST, ISO 27001, PCI DSS.
- Risk assessment and mitigation planning.
- Practical: Conduct a basic risk assessment for a mock organization.

Week 2: Linux and Windows Security Hardening

- Linux: Using ufw, iptables, and SELinux.
- Windows: Configuring the firewall, BitLocker, and Defender.
- Practical: Hardening Linux and Windows environments for security.

Week 3: Bash Scripting and Python for Cybersecurity (Part 1)

- Introduction to Bash scripting for automation.
- Python fundamentals: Loops, conditionals, and file handling.
- Practical: Automate log analysis and backups using Bash and Python.

Week 4: Vulnerability Scanning and Virtualization

- Tools: Nmap, OpenVAS, and Nessus.
- Using VirtualBox/VMware for secure testing environments.
- Practical: Scan and identify vulnerabilities in a sandbox environment.

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Phase 2: Advanced Cybersecurity Concepts (3 Months)

Month 4: Advanced System, Application, and Web Security

Week 1: Advanced Linux Security

- Hardening SSH (key-based authentication, disabling root login).
- File integrity monitoring (Tripwire, AIDE).
- Practical: Secure a Linux server and simulate attacks to test hardening.

Week 2: Advanced Windows Security

- Configuring group policies for security.
- Windows event logs for auditing and monitoring.
- Practical: Create group policies to enforce security and analyze event logs.

Week 3: Web Application Security

- OWASP Top 10 vulnerabilities deep dive.
- Exploiting and mitigating SQL Injection, XSS, CSRF.
- Practical: Perform vulnerability analysis and fix on a sample web app.

Week 4: Database Security

- Introduction to SQL and NoSQL databases.
- Securing database connections, queries, and backups.
- Practical: Harden MySQL and MongoDB databases and monitor activity.

Month 5: Cloud, IoT, and Cryptography Security

Week 1: Cloud Security

- Cloud service models: IaaS, PaaS, SaaS.
- Identity and access management (IAM) in AWS/Azure.
- Practical: Secure an AWS S3 bucket and set up IAM roles.

Week 2: IoT Security

- IoT architecture and common vulnerabilities.
- Securing IoT protocols and devices.
- Practical: Simulate IoT attacks using tools and mitigate vulnerabilities.

Week 3: Cryptography (Advanced)

- Understanding PKI and managing certificates.

- Practical: Configure and test a VPN using OpenVPN.

Week 4: Windows Networking and Security Tools

- Advanced networking tools: PowerShell commands (Test-NetConnection, Get-NetIPAddress).
- Practical: Analyze and secure a Windows-based network.

Month 6: DevOps, Incident Response, and Capstone Project

Week 1: DevOps Security

- Introduction to DevSecOps principles.
- Securing CI/CD pipelines with Jenkins, Docker, and Kubernetes.
- Practical: Build and secure a simple CI/CD pipeline.

Week 2: Incident Response and Forensics

- Incident response lifecycle and creating response plans.
- Introduction to forensic tools: Autopsy, Volatility.
- Practical: Simulate and analyze a forensic investigation.

Week 3: Capstone Project (Part 1)

- Design a secure system architecture for a small organization.
- Implement user access controls and secure communication.

Week 4: Capstone Project (Part 2)

- Identify and mitigate vulnerabilities in the system.
- Final presentation and documentation.

Program Highlights

- **Hands-on Labs:** Each week includes practical labs to reinforce theoretical concepts.
- **Capstone Project:** A real-world project to assess students' understanding and application of skills.
- **Virtualization:** All practical exercises conducted in secure virtualized environments.
- **Focus on Tools:** Students gain experience with industry-standard tools like Wireshark, Metasploit, OpenVAS, Nessus, and Docker.

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Ethical Hacking Course Outline (3 Months)

Phase 1: Basics of Ethical Hacking and Reconnaissance (Month 1)

Week 1: Introduction to Ethical Hacking

- **Concepts and Ethics**:
  - What is ethical hacking? Understanding the hacker mindset.
  - White Hat vs. Black Hat vs. Grey Hat.
  - Legal considerations and certifications (CEH, OSCP, etc.).
- **Hands-on**:
  - Setting up a hacking lab using VirtualBox/VMware with Kali Linux and vulnerable VMs (e.g., Metasploitable, DVWA).
  - Implementing and maintaining anonymity in Linux labs using Proxy-Chains

Week 2: Reconnaissance and Information Gathering

- **Passive Reconnaissance**:
  - Open-source intelligence (OSINT) tools: Maltego, theHarvester.
  - Searching DNS records, WHOIS, and online databases.
- **Active Reconnaissance**:
  - Scanning networks and hosts using Nmap.
  - Banner grabbing and service enumeration.
- **Hands-on**:
  - Perform OSINT using Maltego.
  - Network scanning and mapping with Nmap.

Week 3: Scanning and Vulnerability Assessment

- **Scanning Tools**:
  - Nessus, OpenVAS, Nikto, and Burp Suite.
  - Understanding CVEs and vulnerability databases.
- **Hands-on**:
  - Use Nessus to scan a target system and generate a report.
  - Identify and analyze vulnerabilities in a sample application.

Week 4: Introduction to Exploitation

- **Basic Exploitation Concepts**:
  - Understanding exploits, payloads, and shell access.
  - Introduction to Metasploit Framework.
- **Hands-on**:
  - Exploit a vulnerable service using Metasploit.
  - Gain access to a test system and analyze the impact.

Phase 2: Intermediate Exploitation and Web Application Security (Month 2)

Week 1: Exploitation Techniques

- **Privilege Escalation**:
  - Windows and Linux privilege escalation techniques.
  - Exploiting misconfigurations and weak passwords.
- **Post-Exploitation**:
  - Maintaining access using backdoors.
  - Data exfiltration and lateral movement.
- **Hands-on**:
  - Perform privilege escalation on a vulnerable Linux system.
  - Set up a reverse shell and maintain persistent access.

Week 2: Web Application Hacking (Part 1)

- **Common Web Vulnerabilities**:
  - OWASP Top 10 overview: XSS, SQL Injection, CSRF, etc.
  - Identifying vulnerabilities in web applications.
- **Hands-on**:
  - Exploit SQL Injection vulnerabilities in DVWA.
  - Use Burp Suite to intercept and modify HTTP requests.

Week 3: Web Application Hacking (Part 2)

- **Exploiting Authentication Mechanisms**:
  - Brute-force and credential stuffing attacks.
  - Bypassing login pages and session hijacking.
- **Hands-on**:

- Perform a brute-force attack using Hydra.
- Analyze and fix session vulnerabilities in a web app.

Week 4: Network Security and Wireless Hacking

- **Network Security Fundamentals**:
    - Sniffing traffic using Wireshark and Tcpdump.
    - Introduction to ARP poisoning and MITM attacks.
- **Wireless Security**:
    - WPA2 cracking and WPS vulnerabilities.
    - Using Aircrack-ng for wireless network attacks.
- **Hands-on**:
    - Capture and analyze traffic using Wireshark.
    - Crack a test WPA2-protected wireless network.

Phase 3: Advanced Ethical Hacking Concepts (Month 3)

Week 1: Advanced Exploitation

- **Buffer Overflow Attacks**:
    - Basics of buffer overflows and crafting payloads.
    - Using Immunity Debugger to exploit buffer overflow vulnerabilities.
- **Hands-on**:
    - Exploit a buffer overflow vulnerability in a controlled lab environment.

Week 2: Advanced Web Application Hacking

- **Server-Side Attacks**:
    - Exploiting file upload vulnerabilities.
    - XML External Entity (XXE) injection.
- **Client-Side Attacks**:
    - Social engineering and phishing techniques.
    - Malware delivery using email and USB devices.
- **Hands-on**:
    - Simulate a phishing attack using SET (Social-Engineer Toolkit).
    - Exploit file upload vulnerability in DVWA.

Week 3: Mobile and IoT Hacking

- **Mobile Application Security**:

    - Analyzing APK files and reverse engineering apps.
    - Exploiting insecure data storage and APIs.
- **IoT Security**:
    - Common IoT vulnerabilities.
    - Analyzing and exploiting IoT devices in a simulated lab.
- **Hands-on**:
    - Perform static analysis on an APK file.
    - Simulate IoT exploitation using IoT-specific tools.

Week 4: Penetration Testing and Reporting

- **Conducting a Penetration Test**:
    - Planning and scoping.
    - Writing a penetration test report.
- **Hands-on**:
    - Perform a full penetration test on a target system.
    - Draft a professional report with findings and recommendations.

Key Program Highlights

1. **Practical Focus**:
    - Each week includes lab sessions for hands-on practice.
    - Students set up their own virtual labs for safe testing and learning.
2. **Tools Covered**:
    - Kali Linux, Metasploit, Nmap, Wireshark, Burp Suite, Nessus, Aircrack-ng, Hydra, Maltego, and others.
3. **Capstone Project**:
    - Simulate a real-world penetration test on a complex target environment.
    - Deliver a professional report with remediation strategies.
4. **Learning Path**:
    - Suitable for beginners, with gradual progression to advanced concepts

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Some Tools to Be Introduced

1. **Wireshark** – Network packet analysis.
2. **Nmap/Nikto** – Network scanning and discovery.
3. **Metasploit** – Exploitation framework.
4. **Kali Linux/Parrot OS** – Penetration testing operating system.
5. **OWASP ZAP** – Web vulnerability scanner.
6. **Snort/Suricata** – Intrusion detection systems.
7. **Splunk/Graylog** – Log analysis tools.
8. **Nessus/OpenVAS** – Vulnerability scanners.
9. **Autopsy** – Forensic analysis tool.
10. **aircrack-ng** – Wireless security tool.
11. **John the Ripper** – Password cracking tool